Intelligent Agent

1

1

What is an Agent?

- An agent is an entity that perceives its environment through sensors and acts upon that environment through actuators.
- It performs tasks on behalf of users, like answering questions, generating content, solving problems, etc.

2

2

What is an Intelligent Agent?

- An intelligent agent is an agent that can autonomously make decisions to achieve specific goals, often using some form of reasoning or learning.

- Intelligent Agent Features:
  - ***reactive***
  - ***pro-active***
  - ***social***

3

3

Reactive Behavior

- The agent responds **immediately** to changes in the environment.
- Sense and respond
- **Example**: A **smoke detector system** sounds an alarm as soon as it detects smoke
- **Pros**: Fast response, good for real-time systems
- **Cons**: No planning, can't anticipate future needs

4

4

Pro-active Behavior

- The agent doesn't just react , it **takes initiative** to achieve its goals.
- Plan and act
- **Example**: A **calendar agent** that reminds you to leave early because it anticipates traffic before your meeting.
- **Pros**: Goal-driven, intelligent decision-making.
- **Cons**: Requires more computation and planning

5

5

Social Behavior

- The agent **communicates** and **collaborates** with other agents or users.
- Work with others
- **Example**: In a **multi-agent system**, delivery drones coordinate to cover different zones efficiently.
- **Pros**: Teamwork, coordination, shared goals.
- **Cons**: Needs protocol, trust, and clear communication rules.

6

6

Example

- Imagine a **smart home assistant**:
  - Turns on lights when you enter a room.
  - Suggests adjusting your sleep schedule based on your calendar and past patterns.
  - Syncs with other devices, negotiates energy use with the smart grid.

7

7

Why intelligent agents?

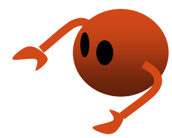
- Handle Complexity and Overload
- Act Autonomously
- Learn and Adapt
- Support Personalization
- Enable Automation in Complex Systems

8

8

What intelligent agents can do for us

- Carry out tasks on the user's behalf
- Train or teach the user
- Help different users collaborate
- Monitor events and procedures



- Specifically, intelligent agents can help us with
  - Information retrieval
  - Information filtering
  - Mail management
  - Recreational activities – selection of books, music, holidays
  - Booking of meetings, hotels, tickets

9

9

What intelligent agents can do for us (cont'd)

Information filtering agent

- One type is the selection of articles from a continuous stream to suit user needs
- User can create "news agents" and train them by giving positive or negative feedback for articles recommended
- The use of key words alone can be restrictive
- Underlying semantics must be extracted for more effectiveness

10

10

What intelligent agents can do for us (cont'd)

Electronic mail agent

- Assist users with electronic mail
- Learn to prioritize, delete, forward, sort and archive mail messages on behalf of the user
- May use intelligent system techniques like case-based reasoning
- Can associate a level of confidence with its action or suggestion
- Use of "do-it" and "tell-me" thresholds set by user
- May involve multi-agent collaboration

11

11

What intelligent agents can do for us (cont'd)

Selection agents for entertainment

- Conversational agents show potential for becoming popular and commercially successful eg ALICE
- Use "social filtering" – correlation between different users to make recommendations on books, CDs, films etc.
- So, if user *A* liked items *X* and *Y*, and user *B* liked item *X* and *Z*, then item *Z* may be recommended for user *A*
- Amazon.com has been using this system for years ->

12

12

Characteristics of a good agent

Action

- Agent must be able to take some action and not just provide advice
- As the Internet becomes more agent-friendly, more capable agents will grow.

13

13

Characteristics of a good agent (cont.)

Autonomy

- An agent can be much more useful if it can act autonomously
- The right level of autonomy for a task must be found

14

14

Characteristics of a good agent (cont.)

Communication

- Must communicate well with the user
- Should understand user's goals, and constraints
- Useful communication requires shared knowledge on
  - language of communication
  - problem domain

15

15

Characteristics of a good agent (cont.)

Adaptation

- Can gain user confidence by learning user preferences
- ML techniques such as ANNS, GAs or CBR can be used
- Adapting to user preferences can be also achieved by using data mining techniques such as clustering
- Agent forms clusters of users with similar features
- User's needs can then be anticipated by placing the user in one of these clusters and analysing the cluster

16

16

Types of agents

- Goal-Based Agents
- Utility-Based Agents
- Learning Agents
- Mobile Agents
- Interface Agents
- Multi-Agent

17

17

Types of agents (cont'd)

- **Goal-Based Agents**
  - Agents that act to achieve specific goals rather than just reacting.
  - **Example:** A GPS app that finds the best route based on a destination.
  - **Pro:** Can handle a wide variety of tasks.
  - **Cons:** Can be computationally expensive due to planning.

18

18

Types of agents (cont'd)

- **Utility-Based Agents**

  - Consider multiple goals and pick actions that maximize a utility function.
  - **Example:** Stock-trading bots balancing risk vs. return.
  - **Pros:** More intelligent decisions; adaptable.
  - **Cons:** Requires well-defined utility models, which can be hard to create.

19

19

Types of agents (cont'd)

- **Learning Agents**

  - Improve their performance over time based on experience.
  - **Example:** Recommendation systems like Netflix or Spotify adapting to your taste.
  - **Pros:** Adaptable to new environments and user preferences.
  - **Cons:** Requires data and time to train effectively.

20

20

Types of agents (cont'd)

- **Mobile Agents**
  - Software agents that can move across different environments.
  - **Example:** An agent that travels to different servers to gather distributed data.
  - **Pros:** Efficient use of network resources.
  - **Cons:** Security concerns.

21

21

Types of agents (cont'd)

- **Interface Agents**
  - Assist users by interacting through a user interface.
  - **Example:** Modern virtual assistants like Siri or Alexa.
  - **Pros:** Improves user productivity and engagement.
  - **Cons:** User acceptance depends on perceived usefulness.

22

22

Types of agents (cont'd)

- **<u>Multi-Agent</u>**
  - Groups of agents working together, possibly with different roles.
  - **Example:** Autonomous drones collaborating in search-and-rescue missions.
  - **Pros:** Scalability, robustness.
  - **Cons:** Coordination and communication complexity.

23

23

Steps to Build an Intelligent Agent

Two main problems to overcome:

- *Competence*
  - How do we build agents with the knowledge needed to decide
    - when to help the user
    - what to help the user with, and
    - how to help the user?
- *Trust*
  - How to guarantee user comfort (and protection!) in delegating tasks to the agent

24

24

Steps to Build an Intelligent Agent (cont’d)

- Step 1: Define the Agent's Purpose
- Step 2: Specify the Environment
- Step 3: Choose the Agent Type
- Step 4: Design the Agent Architecture
- Step 5: Implement the Agent Logic
- Step 6: Train the Agent
- Step 7: Test in Simulated or Real Environment

25

25

Steps to Build an Intelligent Agent (cont’d)

- ***Step 1: Define the Agent's Purpose:***
  - **What task** will the agent perform?
  - **Why** do you need an agent instead of a traditional program?
  - Define the **environment.**

26

26

Steps to Build an Intelligent Agent (cont'd)

- ***Step 2: Specify the Environment:***
  - Percepts (what the agent can sense)
  - Actions (what the agent can do)
  - Goals (what is a "good" outcome)

27

27

Steps to Build an Intelligent Agent (cont'd)

- ***Step 3: Choose the Agent Type***
  - Model-based
  - Goal-based
  - Utility-based
  - Learning agent
  - Mobile/interface/collaborative agents

28

28

Steps to Build an Intelligent Agent (cont'd)

- ***Step 4: Design the Agent Architecture***
  - Perception module (sensors, data input)
  - Decision-making module (rules, logic, planning)
  - Action

29

29

Steps to Build an Intelligent Agent (cont'd)

- ***Step 5: Implement the Agent Logic***
  - Rule-based systems (if...then)
  - Search and planning algorithms
  - Machine learning models.

30

30

Steps to Build an Intelligent Agent (cont'd)

- ***Step 6: Train the Agent***
  - Provide datasets for training
  - Use feedback

31

31

Steps to Build an Intelligent Agent (cont'd)

- ***Step 7: Test***
  - Performance (speed, accuracy)
  - Scalability

32

32

Steps to Build an Intelligent Agent (cont'd)

- ***Step 8: Iterate and Improve***
  - Gather user feedback or data logs
- Refine:
  - Agent knowledge/rules
  - Learning models
  - Decision-making strategies

33

33

Case Study: Building an Intelligent Email Sorting Agent

Create an intelligent agent that helps users **organize and manage their email inbox** by automatically:

- Filtering spam
- Categorizing messages (e.g., work, personal, promotions)
- Prioritizing important emails

34

34

Case Study: Building an Intelligent Email Sorting Agent

- **Define the Problem:**
  - **User pain**:
    - Too many emails, hard to find important one
  - **Agent goal**:
    - Reduce inbox clutter and highlight priority messages

35

35

Case Study: Building an Intelligent Email Sorting Agent

- **Specify the Environment:**
  - **Percepts** :
    - Incoming email data
  - **Action**:
    - Mark email as spam, categorize into folders, mark as high-priority
  - **Performance Measure:**
    - User satisfaction, correct classification rate, time saved

36

36

Case Study: Building an Intelligent Email Sorting Agent

- **Choose the Agent Type**
  - Model-based
  - Utility
  - Learning Agent

37

37

Case Study: Building an Intelligent Email Sorting Agent

- **Design Agent Architecture**
  - **Input Layer**:
    - Email content (subject, sender, body)
  - **Processing Layer**:
    - Spam filter (pre-trained ML model)
    - Classifier (categorizes into folders)
    - Priority scorer (based on sender, keywords, history)
  - **Output Layer**:
    - Email is tagged, moved, or prioritized in inbox

38

38

Case Study: Building an Intelligent Email Sorting Agent

- **Implement Core Logic**
  - **Use** :
    - **Naive Bayes** for spam detection
    - **SVM** for classification
    - **Decision tree or NN** to assign priority

39

39

Case Study: Building an Intelligent Email Sorting Agent

- **Enable Learning**
  - **Feedback loop:**
    - User can "mark as important" or "not spam"
  - Agent uses supervised learning to improve accuracy over time

40

40

Case Study: Building an Intelligent Email Sorting Agent

- **Testing**
  - Train on a labeled dataset
  - Test classification accuracy and user feedback
  - Run A/B test with users to compare inbox efficiency before/after agent

41

41

Case Study: Building an Intelligent Email Sorting Agent

- **Iterate and Improve**
  - Time-sensitive alerts
  - Suggested replies
  - Learning user writing style

42

42

Case Study: Building an Intelligent Email Sorting Agent

- **Agent Characteristics in Action:**
  - **Reactive**:
    - Filters spam as emails arrive
  - **Pro-active**:
    - Highlights urgent emails before user opens inbox
  - **Social**:
    - Learns from user feedback and adapts behavior